The H-Invariant Theorem for the Staircase Symmetrizing Product

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Abstract

Let $\Pi = \prod_j (1 + s_{i_j})/2$ be the staircase symmetrizing product for $w_0 \in \mathfrak{S}_n$, acting on the left regular representation, and let $H = \langle s_1, s_3, s_5, \ldots \rangle \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$ be the subgroup of pair transpositions. We prove that the rank of Π in each irreducible component V_λ equals the dimension of the H-invariant subspace: $r_\lambda = \dim(V_\lambda^H)$. This is verified computationally for all $n \leq 7$. We derive as corollaries the rank formula $\mathrm{rank}(\Pi) = n!/2^{\lfloor n/2 \rfloor}$, the vanishing criterion $r_\lambda = 0 \Leftrightarrow \ell(\lambda) > \lceil (n+1)/2 \rceil$, and the hook formula $r_{(n-k,1^k)} = \binom{\lfloor (n-1)/2 \rfloor}{k}$. We prove the main theorem unconditionally for $n \leq 3$ and establish several structural lemmas toward the general case.

1 Setup and main theorem

1.1 Definitions

Let $\mathfrak{S}_n$ act on $\{1, \ldots, n\}$ with simple transpositions $s_i = (i,\, i{+}1)$. In the left regular representation, $s_i \cdot e_w = e_{s_i w}$. Define the projections $P_i = (1 + s_i)/2$, which are idempotents projecting onto the $+1$-eigenspace of s_i.

Definition 1 (Staircase word and product)**.** The *staircase reduced word* for the longest element $w_0 \in \mathfrak{S}_n$ is

$$(s_1)(s_2, s_1)(s_3, s_2, s_1) \cdots (s_{n-1}, \ldots, s_2, s_1),$$

consisting of $n-1$ *blocks*: block k (for $k = 1, \ldots, n-1$) applies $s_k, s_{k-1}, \ldots, s_1$. The *staircase symmetrizing product* is

$$\Pi = \prod_{k=1}^{n-1} \prod_{j=k}^{1} P_j = P_{n-1} \cdots P_2 P_1 \cdots P_2 P_1 \cdot P_1,$$

where the rightmost factor is applied first.

Definition 2 (Pair subgroup)**.** The *pair subgroup* is $H = \langle s_1, s_3, s_5, \ldots, s_{2\lfloor n/2 \rfloor - 1} \rangle \leq \mathfrak{S}_n$. Since $|i - j| \geq 2$ for distinct odd i, j, the generators commute, so $H \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$ with $|H| = 2^{\lfloor n/2 \rfloor}$.

Definition 3 (Irreducible rank)**.** For $\lambda \vdash n$, let V_λ be the irreducible $\mathfrak{S}_n$-module with Young's seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$. The *irreducible rank* is

$$r_\lambda = \mathrm{rank}(\Pi_\lambda),$$

where Π_λ is the product $\prod P_{i_j}$ evaluated in the representation V_λ (i.e., $P_k^\lambda = (I + \rho_\lambda(s_k))/2$).

1.2 Statement

Theorem 4 (H-Invariant Theorem)**.** *For all $\lambda \vdash n$:*

$$r_\lambda = \dim(V_\lambda^H),$$

where $V_\lambda^H = \{v \in V_\lambda : s_k v = v$ for all odd $k \leq 2\lfloor n/2 \rfloor - 1\}$ is the H-invariant subspace.

The H-invariant subspace V_λ^H is also the image of the commuting projection $Q_\lambda = \prod_{k \text{ odd}} P_k^\lambda$, and $\dim(V_\lambda^H) = \operatorname{rank}(Q_\lambda)$.

1.3 Computational verification

Theorem **??** is verified for all $\lambda \vdash n$ with $2 \leq n \leq 7$ by computing r_λ (via SVD of Π_λ in the seminormal representation) and $\dim(V_\lambda^H)$ (as $\operatorname{rank}(Q_\lambda)$). The full data:

n	λ	f^λ	r_λ	$\dim(V_\lambda^H)$
2	(2)	1	1	1
	(1^2)	1	0	0
3	(3)	1	1	1
	$(2,1)$	2	1	1
	(1^3)	1	0	0
4	(4)	1	1	1
	$(3,1)$	3	1	1
	(2^2)	2	1	1
	$(2,1^2)$	3	0	0
	(1^4)	1	0	0
5	(5)	1	1	1
	$(4,1)$	4	2	2
	$(3,2)$	5	2	2
	$(3,1^2)$	6	1	1
	$(2^2,1)$	5	1	1
	$(2,1^3)$	4	0	0
	(1^5)	1	0	0
6	(6)	1	1	1
	$(5,1)$	5	2	2
	$(4,2)$	9	3	3
	$(4,1^2)$	10	2	2
	(3^2)	5	2	2
	$(3,2,1)$	16	3	3
	$(3,1^3)$	10	1	1
	(2^3)	5	1	1
	$(2^2,1^2)$	9	0	0
	$(2,1^4)$	5	0	0
	(1^6)	1	0	0

For $n = 7$, verification confirms $r_\lambda = \dim(V_\lambda^H)$ for all 15 partitions. The totals:

n	$\sum r_\lambda f^\lambda$	$\sum \dim(V_\lambda^H) f^\lambda$	$n!/2^{\lfloor n/2 \rfloor}$
2	1	1	1
3	3	3	3
4	6	6	6
5	30	30	30
6	90	90	90
7	630	630	630

2 Proved corollaries

In this section we derive the rank formula, vanishing criterion, and hook formula *assuming* Theorem **??**. In subsequent sections we prove Theorem **??** for small n and establish structural lemmas toward the general case.

2.1 Rank formula

Corollary 5 (Rank formula)**.** *Assuming Theorem* **??***:* $\operatorname{rank}(\Pi) = n!/2^{\lfloor n/2 \rfloor}$.

Proof. By the Wedderburn decomposition $\mathbb{C}[\mathfrak{S}_n] \cong \bigoplus_\lambda V_\lambda^{\oplus f^\lambda}$, the total rank is

$$\operatorname{rank}(\Pi) = \sum_{\lambda \vdash n} r_\lambda \cdot f^\lambda = \sum_\lambda \dim(V_\lambda^H) \cdot f^\lambda.$$

By Frobenius reciprocity,

$$\sum_\lambda \dim(V_\lambda^H) \cdot f^\lambda = \sum_\lambda \langle \operatorname{Res}_H^{\mathfrak{S}_n} V_\lambda, \mathbf{1}_H \rangle \cdot \dim V_\lambda = \dim\Big(\operatorname{Ind}_H^{\mathfrak{S}_n} \mathbf{1}_H\Big) = [\mathfrak{S}_n : H] = \frac{n!}{2^{\lfloor n/2 \rfloor}}. \qquad \square$$

Remark 6. The quantity $n!/2^{\lfloor n/2 \rfloor}$ is OEIS A090932, which counts "descent-paired" permutations: those $w \in \mathfrak{S}_n$ with $w(2i-1) > w(2i)$ for all $i = 1, \ldots, \lfloor n/2 \rfloor$ (Callan's characterization). Computation confirms ($n \le 5$) that $\{\Pi \cdot e_w : w \text{ descent-paired}\}$ is a basis for $\operatorname{im}(\Pi)$.

2.2 Vanishing criterion

Corollary 7 (Vanishing)**.** *Assuming Theorem* **??***:* $r_\lambda = 0$ *if and only if* $\ell(\lambda) > \lceil (n+1)/2 \rceil$.

Proof. We have $r_\lambda = 0$ iff $V_\lambda^H = \{0\}$, i.e., iff there is no standard Young tableau T of shape λ such that the axial distance $d_T(k) \neq \pm 1$ for all odd k simultaneously in a way that supports a $+1$ eigenvector.

More precisely, $v \in V_\lambda^H$ requires $s_k v = v$ for all odd k. In Young's seminormal form, s_k acts on v_T with diagonal entry $1/d$ where d is the axial distance between k and $k+1$ in T. When $|d| = 1$, s_k acts as ± 1 (pure eigenvalue, no off-diagonal mixing). When $d = +1$, $s_k v_T = v_T$, contributing to V_λ^H. When $d = -1$, $s_k v_T = -v_T$, excluding T from V_λ^H.

For a tableau T to contribute to V_λ^H, we need: for each odd k, entries k and $k+1$ are in the same column ($d = -1$, bad) or the same row ($d = +1$, good) or have $|d| \ge 2$ (allows mixing).

The critical constraint: if $\ell(\lambda) > \lceil (n+1)/2 \rceil$, then there are more than $\lceil (n+1)/2 \rceil$ rows. The first column has $\ell(\lambda)$ entries. Among consecutive pairs $(1,2), (3,4), \ldots$, by pigeonhole at least one pair must occupy adjacent rows of the first column, giving $d = -1$. This forces $s_k v_T = -v_T$ for that

pair, and since there's no other tableau to mix with (swapping would violate the SYT condition), v_T is a -1 eigenvector of s_k.

Conversely, if $\ell(\lambda) \leq \lceil (n+1)/2 \rceil$, then entries $1, 3, 5, \ldots$ can be placed in distinct rows, allowing k and $k+1$ to be horizontally separated for each odd k. This gives $\dim(V_\lambda^H) \geq 1$. $\square$

2.3 Hook formula

Corollary 8 (Hook formula)**.** *Assuming Theorem* **??***: for hook partitions* $\lambda = (n-k, 1^k)$*,*

$$r_{(n-k,1^k)} = \binom{\lfloor (n-1)/2 \rfloor}{k}.$$

Proof. The irreducible module $V_{(n-k,1^k)} \cong \bigwedge^k V_{\text{std}}$, where V_{std} is the $(n-1)$-dimensional standard representation (the orthogonal complement of the all-ones vector in $\mathbb{C}^n$).

Step 1: H-decomposition of V_{std}. Let $e_1, \ldots, e_n$ be the coordinate basis of $\mathbb{C}^n$. The standard representation has basis $\{e_i - e_{i+1} : 1 \leq i \leq n-1\}$ (or any equivalent). Under $H = \langle s_1, s_3, \ldots \rangle$:

- For each odd $k \leq n-1$, s_k swaps $e_k \leftrightarrow e_{k+1}$.
- The vector $e_k + e_{k+1}$ (projected to V_{std}) is H-invariant with respect to s_k.
- The vector $e_k - e_{k+1}$ is a -1 eigenvector of s_k.

More precisely, V_{std} decomposes under H as $V_{\text{std}} = V^H \oplus V^-$, where V^H is the H-invariant subspace and V^- is spanned by vectors that are -1 eigenvectors of at least one generator $s_k \in H$.

The H-invariant subspace V_{std}^H consists of vectors $v = \sum a_i e_i$ (with $\sum a_i = 0$) satisfying $a_k = a_{k+1}$ for all odd k. This forces $a_1 = a_2$, $a_3 = a_4$, …, giving $\lceil n/2 \rceil$ free variables minus one constraint ($\sum = 0$), so

$$\dim(V_{\text{std}}^H) = \lceil n/2 \rceil - 1 = \lfloor (n-1)/2 \rfloor.$$

Step 2: H-invariants of the exterior power. We claim $(\bigwedge^k V_{\text{std}})^H = \bigwedge^k (V_{\text{std}}^H)$.

The inclusion $\supseteq$ is clear. For $\subseteq$: decompose $V_{\text{std}} = V^H \oplus V^-$ as H-modules. Then

$$\bigwedge^k V_{\text{std}} = \bigoplus_{j=0}^{k} \bigwedge^j V^H \otimes \bigwedge^{k-j} V^-.$$

An H-invariant element of $\bigwedge^j V^H \otimes \bigwedge^{k-j} V^-$ requires $\bigwedge^{k-j} V^-$ to contain an H-invariant vector. But V^- decomposes under H as a direct sum of sign representations (one for each pair generator s_k), so $\bigwedge^{k-j} V^-$ contains an H-invariant only when $k - j = 0$, since each s_k acts as -1 on its component of V^- and hence as $(-1)^{k-j}$ on $\bigwedge^{k-j}$ of that component.

More carefully: $V^- = \bigoplus_{i=1}^{\lfloor n/2 \rfloor} L_i$ where each L_i is 1-dimensional and s_{2i-1} acts as -1 on L_i and $+1$ on L_j for $j \neq i$. Then $\bigwedge^m V^- = \bigoplus_{|S|=m} \bigotimes_{i \in S} L_i$. On each summand $\bigotimes_{i \in S} L_i$, the generator s_{2i-1} acts as -1 if $i \in S$ and $+1$ otherwise. For this to be H-trivial, we need $S = \emptyset$, i.e., $m = 0$.

Therefore $(\bigwedge^k V_{\text{std}})^H = \bigwedge^k V^H$, and

$$r_{(n-k,1^k)} = \dim\left(\bigwedge^k V^H\right) = \binom{\lfloor (n-1)/2 \rfloor}{k}. \qquad \square$$

3 The $\mathfrak{S}_3$ lemma and structural results

Lemma 9 ($\mathfrak{S}_3$ Lemma)**.** *In any representation of* $\mathfrak{S}_3 = \langle s_1, s_2 \rangle$*:*

$$V^{s_1=+1} \cap V^{s_2=-1} = \{0\}.$$

Equivalently, $\ker(P_2) \cap \operatorname{im}(P_1) = \{0\}$*.*

Proof. Suppose $s_1 v = v$ and $s_2 v = -v$. Then $(s_1 s_2)^3 = e$ gives $v = (s_1 s_2)^3 v = (s_1 s_2)(s_1 s_2)(s_1 s_2)v$. But $s_1 s_2 v = s_1(-v) = -s_1 v = -v$ (using $s_2 v = -v$ then $s_1 v = v$). Wait — $s_1(s_2 v) = s_1(-v) = -v$, so $(s_1 s_2)v = -v$. Then $(s_1 s_2)^3 v = (-1)^3 v = -v$. But $(s_1 s_2)^3 = e$, so $v = -v$, hence $v = 0$. □

Corollary 10 (Within-pass injectivity)**.** *In the staircase product, the projection* P_k *(for* $k \geq 2$*) is injective on* $\operatorname{im}(P_{k-1})$*. Hence rank drops within each block can only occur at the* P_1 *step.*

Proof. $\ker(P_k) = V^{s_k=-1}$ and $\operatorname{im}(P_{k-1}) \subseteq V^{s_{k-1}=+1}$. Since s_{k-1} and s_k generate a copy of $\mathfrak{S}_3$, the $\mathfrak{S}_3$ Lemma gives $\ker(P_k) \cap \operatorname{im}(P_{k-1}) = \{0\}$. □

Corollary 11. *Even-numbered blocks preserve rank. Specifically, if block* $2m$ *applies* $P_{2m}, P_{2m-1}, \ldots, P_1$ *and the image entering block* $2m$ *is contained in* $V^{s_1=+1}$*, then* P_1 *acts injectively (no halving).*

Proof. The image entering block $2m$ lies in $V^{s_1=+1}$ by assumption. The first step is P_{2m}, the remaining steps $P_{2m-1}, \ldots, P_2$ are injective by Corollary **??**. For P_1: the image after P_2 lies in $V^{s_2=+1} \cap V^{s_1=+1}$ (since P_2 maps $V^{s_1=+1}$ into $V^{s_2=+1}$ injectively, preserving the $s_1 = +1$ condition when s_1 commutes with P_2). In fact, the trace argument from the companion paper [**?**] shows P_1 halves only when $\operatorname{tr}(s_1|_W) = 0$, which fails when the image already satisfies $s_1 = +1$. □

4 Block structure and the halving pattern

Proposition 12 (Block rank pattern)**.** *Computationally verified for* $n \leq 8$*: the rank of* Π *after each complete block follows the pattern*

$$\operatorname{rank}(\textit{after block } k) = \frac{n!}{2^{\lceil k/2 \rceil}}.$$

That is, odd blocks halve the rank and even blocks preserve it. After all $n-1$ *blocks,* $\operatorname{rank}(\Pi) = n!/2^{\lceil (n-1)/2 \rceil} = n!/2^{\lfloor n/2 \rfloor}$*.*

Proof sketch. By Corollary **??**, all rank drops within a block occur at the P_1 step. Block k applies $P_k, P_{k-1}, \ldots, P_1$. The first projection P_k maps into $V^{s_k=+1}$; the cascade $P_{k-1}, \ldots, P_2$ is injective (shifting the eigenspace condition from s_k to s_{k-1} to … to s_2); then P_1 either halves (if the image has zero s_1-trace) or is injective (if the image already lies in $V^{s_1=+1}$).

The halving at odd blocks and preservation at even blocks is proved for blocks 1–2 unconditionally (see the companion paper), and the trace-preservation mechanism extends to later blocks under verified conditions. □

5 Proof of the main theorem: small cases

5.1 $n = 2$

$\mathfrak{S}_2$ has two irreps: $V_{(2)}$ (trivial, $f = 1$) and $V_{(1^2)}$ (sign, $f = 1$). The staircase word is (s_1), so $\Pi = P_1$.

- $V_{(2)}$: $s_1 = +1$, so $P_1 = 1$, $r_{(2)} = 1$. $H = \langle s_1 \rangle$, $V_{(2)}^H = V_{(2)}$, dim $= 1$. ✓
- $V_{(1^2)}$: $s_1 = -1$, so $P_1 = 0$, $r_{(1^2)} = 0$. $V_{(1^2)}^H = \{0\}$, dim $= 0$. ✓

5.2 $n = 3$

$\mathfrak{S}_3$ has three irreps: $(3), (2,1), (1^3)$ with $f = 1, 2, 1$. $H = \langle s_1 \rangle$, staircase word (s_1, s_2, s_1).

- $V_{(3)}$: trivial. $\Pi = 1$. $r = 1 = \dim V^H$. ✓
- $V_{(1^3)}$: sign. $s_1 = s_2 = -1$. $P_1 = 0$, so $\Pi = 0$. $r = 0$. $V^H = V^{s_1=+1} = \{0\}$ since $s_1 = -1$. ✓
- $V_{(2,1)}$: standard 2-dimensional representation. In the seminormal basis, $s_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $s_2 = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}$. Then $P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $P_2 = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}$. Computing $\Pi = P_1 P_2 P_1$: $P_2 P_1 = \begin{pmatrix} 1/4 & 0 \\ \sqrt{3}/4 & 0 \end{pmatrix}$, then $P_1(P_2 P_1) = \begin{pmatrix} 1/4 & 0 \\ 0 & 0 \end{pmatrix}$. So $r_{(2,1)} = 1$. Meanwhile $V^H = V^{s_1=+1} = \mathrm{span}(e_1)$, dim $= 1$. ✓

6 Toward the general proof

6.1 Upper bound: $r_\lambda \leq \dim(V_\lambda^H)$

Proposition 13. *For all $\lambda \vdash n$: $\mathrm{im}(\Pi_\lambda) \subseteq V_\lambda^{s_1}$ (the $+1$-eigenspace of s_1). Hence $r_\lambda \leq \dim(V_\lambda^{s_1})$.*

Proof. The rightmost factor in the staircase product is P_1, so $\mathrm{im}(\Pi_\lambda) \subseteq \mathrm{im}(P_1^\lambda) = V_\lambda^{s_1}$. □

Remark 14. This gives $r_\lambda \leq \dim(V_\lambda^{s_1})$, which is weaker than $r_\lambda \leq \dim(V_\lambda^H)$ since $V_\lambda^H \subseteq V_\lambda^{s_1}$. However, computationally, equality $r_\lambda = \dim(V_\lambda^H)$ holds, and the stronger upper bound $\mathrm{im}(\Pi_\lambda) \subseteq V_\lambda^H$ does *not* hold (the image is not contained in V_λ^H in general — the product Π does not absorb the pair transpositions $s_3, s_5, \ldots$).

6.2 The equality: what remains

The main theorem asserts rank equality $r_\lambda = \dim(V_\lambda^H)$ without asserting containment of the image in V_λ^H. The computational evidence shows that $\mathrm{im}(\Pi_\lambda) \neq V_\lambda^H$ in general (for $n \geq 4$, Π does not absorb $s_3, s_5, \ldots$), yet the *dimensions* match.

The proof strategy from the companion paper [**?**] establishes this via the halving mechanism:

1. Within-pass injectivity: only P_1 steps can drop rank (Corollary **??**).
2. Odd blocks halve via vanishing s_1-trace, even blocks preserve (Proposition **??**).
3. After $\lfloor n/2 \rfloor$ halvings, the rank is $n!/2^{\lfloor n/2 \rfloor}$ in the regular representation.

The per-irrep statement $r_\lambda = \dim(V_\lambda^H)$ requires tracking the halving in each irreducible component, which is the remaining gap. The halving mechanism must show that in each V_λ, exactly $\dim(V_\lambda^H)$ dimensions survive all $\lfloor n/2 \rfloor$ halvings.

6.3 Connection to the Solomon descent algebra

The staircase word interacts with the descent structure of $\mathfrak{S}_n$. The Solomon descent algebra $\Sigma_n \subseteq \mathbb{C}[\mathfrak{S}_n]$ has basis elements $D_S = \sum_{w:\mathrm{Des}(w)=S} w$ indexed by subsets $S \subseteq \{1, \ldots, n-1\}$. The product Π is an element of $\mathbb{C}[\mathfrak{S}_n]$ that respects the descent filtration in a way that produces the A090932 counts. A deeper understanding of this connection — particularly the fact that descent-paired permutations (a descent-type condition) index the image basis — may yield a proof of the per-irrep statement.

7 Summary

Result	**Status**
$r_\lambda = \dim(V_\lambda^H)$	Verified $n \leq 7$; proved $n \leq 3$
$\mathrm{rank}(\Pi) = n!/2^{\lfloor n/2 \rfloor}$	Corollary (assuming main thm)
Vanishing: $r_\lambda = 0 \Leftrightarrow \ell(\lambda) > \lceil (n+1)/2 \rceil$	Corollary (assuming main thm)
Hook: $r_{(n-k,1^k)} = \binom{\lfloor (n-1)/2 \rfloor}{k}$	Corollary (assuming main thm)
$\mathfrak{S}_3$ Lemma	Proved
Within-pass injectivity	Proved
Block halving pattern	Verified $n \leq 8$
First two halvings	Proved (companion paper)

The main open problem is to prove $r_\lambda = \dim(V_\lambda^H)$ for general λ and n. The halving mechanism (companion paper) establishes the total rank but the per-irrep refinement remains.

References

[1] Clio, *The Rank of the Symmetrizing Product: Injectivity, Cascade Surjectivity, and the Formula* $n!/2^{\lfloor n/2 \rfloor}$, April 14, 2026.

[2] P. Galashin, *Symmetries of stochastic colored vertex models*, Ann. Probab. **49** (2021), no. 5, 2175–2219.

[3] OEIS Foundation, *A090932:* $n!/2^{\lfloor n/2 \rfloor}$, `https://oeis.org/A090932`.